Principles of Economics

Final term

Topic 1: Measuring National Income - GDP

Objectives and Instruments of Macroeconomics

- Objectives:

Output: High level and rapid growth of output

Employment: High level of employment with involuntary unemployment

Price-level stability

Instruments of Macroeconomics

- Instruments
- Monetary policy:

  Controlling the money supply to determine the interest rates

- Fiscal Policy:

  Government expenditure

  Taxation

Measuring National Income

- *Three main statistics of macroeconomics*:

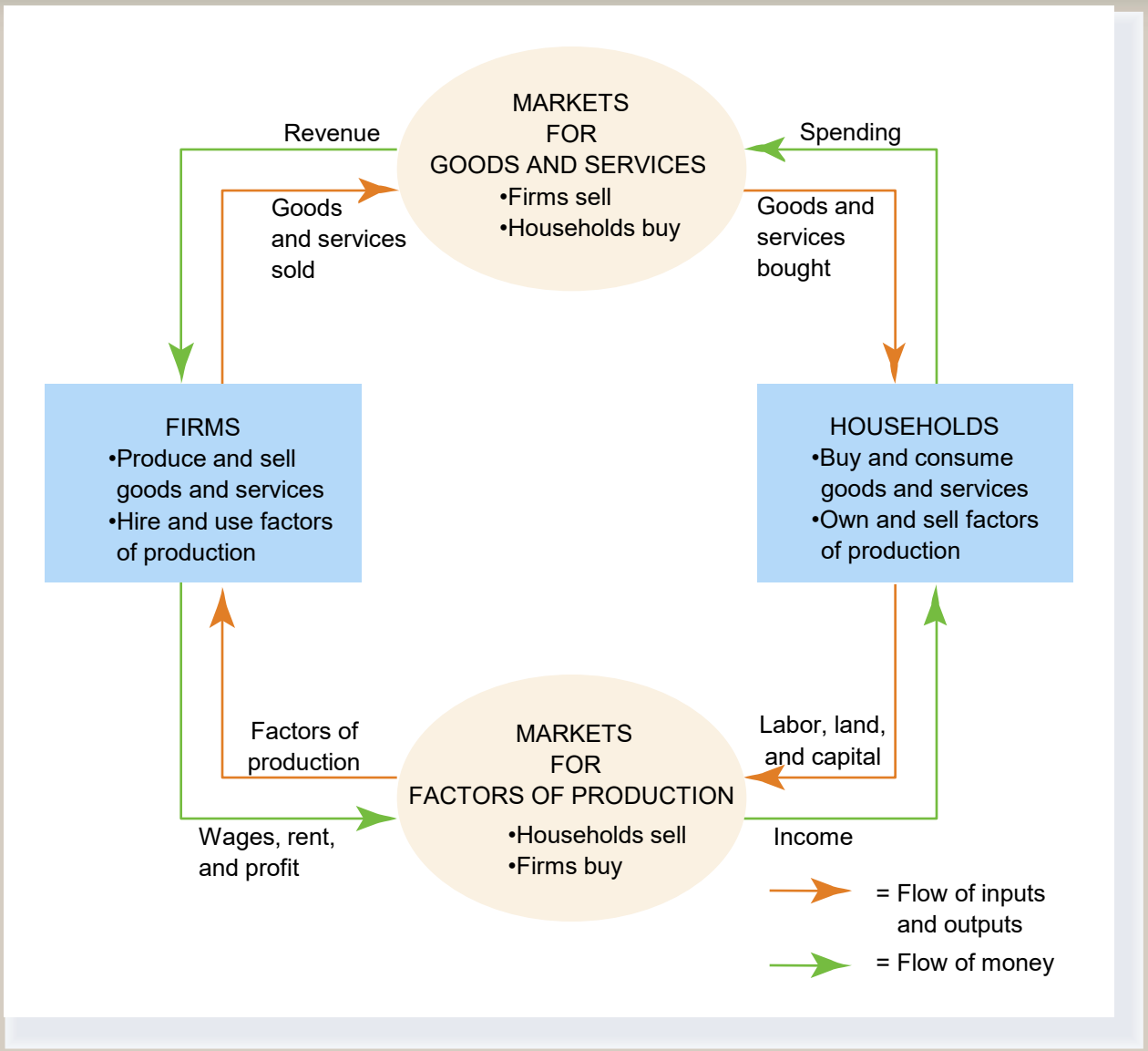
1. **GDP**—Gross domestic product, which tells us the nation's total income and the total expenditure on its output of goods and services
2. **CPI**—the Consumer Price Index, which measures the level of prices (inflation)
3. **Unemployment rate** —this tells the fraction of workers who are unemployed.

Economic Models

- Economists use models to simplify reality in order to improve our understanding of the world.

- The most basic macroeconomic model include:
  - The Circular Flow Diagram

Figure 1 The Circular-Flow Diagram



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The Circular-Flow Diagram

- The *circular-flow diagram* is a visual model of the economy that shows how money flow through markets among households and firms.

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- *Gross domestic product (GDP)* is a measure of the income and expenditures of an economy.

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- GDP is the market value of all final goods and services produced within a country in a given period of time.

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- "GDP is the Market Value . . ."
  - Output is valued at market prices.
- ". . . Of All Final . . ."
  - It records only the value of final goods, not intermediate goods (the value is counted only once).
  - Hallmark Card: Paper is an intermediate good.
- ". . . Goods and Services . . . "
  - It includes both tangible goods (food, clothing, cars) and intangible services (haircuts, housecleaning, doctor visits).

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- “. . . Produced . . .”
  - It includes goods and services currently produced, not transactions involving goods produced in the past.
- “ . . . Within a Country . . .”
  - It measures the value of production within the geographic confines of a country.

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- "... In a Given Period of Time."
  - It measures the value of production that takes place within a specific interval of time, usually a year or a quarter.

- GDP includes all items produced in the economy and sold *legally* in markets.

REAL VERSUS NOMINAL GDP

- *Nominal GDP* values the production of goods and services at *current prices*.

Year	Calculating Nominal GDP
2001	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2002	($2 per hot dog × 150 hot dogs) + ($3 per hamburger × 100 hamburgers) = $600
2003	($3 per hot dog × 200 hot dogs) + ($4 per hamburger × 150 hamburgers) = $1,200

REAL VERSUS NOMINAL GDP

- *Real GDP* values the production of goods and services at *constant prices*.

Year	Calculating Real GDP (base year 2001)
2001	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2002	($1 per hot dog × 150 hot dogs) + ($2 per hamburger × 100 hamburgers) = $350
2003	($1 per hot dog × 200 hot dogs) + ($2 per hamburger × 150 hamburgers) = $500

Not Counted in GDP

- What Is Not Counted in GDP?
  - GDP excludes most items that are produced and consumed at home and that never enter the marketplace.
  - It excludes items produced and sold illicitly, such as illegal drugs.

THE COMPONENTS OF GDP

- GDP (*Y*) is the sum of the following:
  - Consumption *(C)*
  - Investment *(I)*
  - Government Purchases *(G)*
  - Net Exports *(NX)*

$$Y = C + I + G + NX$$

THE COMPONENTS OF GDP

- *Consumption (C)*:
  - The spending by households on goods and services, with the exception of purchases of new housing.
- *Investment (I)*:
  - The spending on capital equipment, inventories, and structures, including new housing.

THE COMPONENTS OF GDP

- *Government Purchases (G)*:
  - The spending on goods and services by local, state, and federal governments.
  - Does *not* include transfer payments because they are not made in exchange for currently produced goods or services.
- *Net Exports (NX):*
  - Exports minus imports.

GDP & Components of Expenditure

GDP

Consumption

- Nondurable goods
- Durable goods
- Services

Investment

- Nonresidential fixed investment
- Residential fixed investment

Government Purchases

- Defense
- Non-defense

Net Export

- Export
- Import

GDP deflator

- GDP deflator comes from nominal and real GDP
- **GDP deflator : Nominal GDP/Real GDP *100**
- With GDP deflator we can also measure the difference in price levels of good and services in a country at a given timeframe

GDP AND ECONOMIC WELL-BEING

- GDP is the best single measure of the economic well-being of a society.
- GDP per person tells us the income and expenditure of the average person in the economy.
- Higher GDP per person indicates a higher standard of living.

GDP AND ECONOMIC WELL-BEING

- GDP is not a perfect measure.
- Some things that contribute to well-being are not included in GDP.
  - The value of leisure.
  - The value of a clean environment.
  - The value of almost all activity that takes place outside of markets, such as the value of the time parents spend with their children and the value of volunteer work.

Summary

- Because every transaction has a buyer and a seller, the total expenditure in the economy must equal the total income in the economy.
- Gross Domestic Product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services.

Summary

- GDP is the market value of all final goods and services produced within a country in a given period of time.
- GDP is divided among four components of expenditure: consumption, investment, government purchases, and net exports.

Summary

- Nominal GDP uses current prices to value the economy's production. Real GDP uses constant base-year prices to value the economy's production of goods and services.

Summary

- GDP is a good measure of economic well-being because people prefer higher to lower incomes.
- It is not a perfect measure of well-being because some things, such as leisure time and a clean environment, aren't measured by GDP.